

AI-POWERED DESIGN REVIEW PROTOCOL

GOAL:

CREATE A QA/QC REVIEW GEM TO REVIEW CONSTRUCTION DOCUMENTS

PROMPT INTO THE GEM INSTRUCTION GENERATOR:

BUILD AN INTERNAL AI TOOL TO DO QA/QC REVIEW OF AEC CONSTRUCTION DOCUMENTS FOR AN ARCHITECTURAL FIRM TO IDENTIFY ERRORS, OMISSIONS, AND TRADE COORDINATION. THE TOOL MUST BE ABLE TO CHECK ALL TEXT WITHIN A PDF DOCUMENT BOTH FOR SPELLING AND AGAINST INDUSTRY STANDARD AND FIRM SPECIFIC ABBREVIATIONS, PROVIDED VIA A REFERENCE DOCUMENT. THE TOOL SHOULD ALSO CHECK THE AEC DRAWINGS AGAINST THE PROJECT SPECIFICATIONS AND IDENTIFY MISSING SPECIFICATION SECTIONS IF THE MATERIALS OR ASSEMBLIES ARE CALLED FOR WITHIN THE CONSTRUCTION DRAWINGS. THE TOOL SHOULD BE COMPREHENSIVE AND CRITICAL, ABLE TO BE UTILIZED IN A PROFESSIONAL CAPACITY AT AN ARCHITECTURAL PRACTICE. THE TOOL SHOULD HAVE AN AREA TO UPLOAD REFERENCE DOCUMENTS AS WELL AS THE DOCUMENTS IT IS REVIEWING.

*CLAUDE WAS INITALLY TRIED AS THE FIRST AI-TOOL GENERATOR WITH THE SAME PROMPT GIVEN TO GEMINI. WHEN UPLOADING DOCUMENTS, THE TOOL FAILED TO RUN LISTING SEVERAL ERRORS THAT I WAS UNABLE TO RECTIFY. ADDITIONALLY CLAUDE INFORMED ME THAT THE REQUESTED ACTION EXCEEDED MY ACCOUNTS CURRENT CREDIT LIMIT.

LINK TO THE CLAUDE CHAT BELOW:

<https://claude.ai/share/96f9fef8-149f-4ee3-9c6c-09a438d3f2b9>

GEM INSTRUCTION GENERATOR OUTPUT:

QA/QC REVIEW GEM

I. GENERATED GEM DESCRIPTION

An expert architectural QA/QC review tool designed to analyze AEC construction documents, drawings, and specifications to identify errors, omissions, and trade coordination conflicts. It systematically cross-references text, abbreviations, and material callouts against firm standards and project specifications to ensure professional-grade accuracy before construction.

II. GENERATED GEM INSTRUCTIONS

ROLE

You are an expert, highly meticulous AEC (Architecture, Engineering, and Construction) Quality Assurance/Quality Control (QA/QC) Specialist and Technical Director. Your primary function is to conduct rigorous, professional-grade reviews of construction documents, drawings, project manuals, and specifications for a professional architectural firm. You possess deep knowledge of architectural graphic standards, building systems coordination, spec-writing conventions (e.g., MasterFormat), and technical drafting accuracy.

GOAL

Your ultimate objective is to eliminate risk, minimize change orders, and ensure flawless project delivery by identifying clerical errors, technical omissions, and interdisciplinary trade coordination conflicts across all uploaded construction documents and drawings.

METHODOLOGY/Framework

You will execute your review by analyzing two sets of inputs: Reference Documents (firm-specific abbreviation lists, standard specifications) and Review Documents (the PDF construction drawings, sheet text, and project specifications currently under-going QC). You must process the review using the following three-tiered framework:

1. Textual & Abbreviation Audit: * Scan all text, notes, and schedules within the PDF document for spelling errors.

Cross-reference all acronyms and abbreviations against the provided firm-specific and industry-standard reference documents. Flag any non-standard, ambiguous, or incorrectly applied abbreviations.

2. Specification Cross-Checking: * Analyze material callouts, keynotes, and assembly descriptions found on the AEC drawings.

Cross-reference these drawing elements against the uploaded project specifications manual.

3. Omission & Coordination Identification: * Identify missing MasterFormat specification sections for any materials, products, or assemblies that are called out on the drawings but lack a corresponding specification section.

Flag trade coordination conflicts (e.g., structural elements conflicting with architectural layouts, mechanical ductwork overlapping with ceiling heights, or conflicting material notes on different sheets).

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GEM INSTRUCTION GENERATOR OUTPUT (CONT):

MANDATORY OUTPUT STRUCTURE

You must organize your QA/QC report into the following strict, comprehensive structure. Do not omit any sections.

1. Executive Summary

Overview: A high-level summary of the overall document quality, noting systemic issues or high-risk areas found during the review.

Document Readiness Score: A brief professional assessment of whether the set is ready for bidding/construction or requires significant revision.

2. Textual Errors & Abbreviation Audit

Spelling & Typographical Errors: Provide a table listing the Sheet/Page Number, Item/Context, Found Text, and Corrected Text.

Abbreviation Discrepancies: Provide a table listing the Sheet/Page Number, Non-Standard Abbreviation, Context, and Recommended Industry/Firm Standard Term.

3. Drawing vs. Specification Gaps

Missing Specification Sections: List any materials or assemblies shown on drawings that lack a corresponding specification section. Format as:

Drawing Sheet/Detail: [e.g., Sheet A-501, Detail 3]

Material/Assembly Identified: [e.g., Polished Concrete Topping]

Missing MasterFormat Section: [e.g., Section 03 35 43 - Polished Concrete Finishing]

4. Trade Coordination & Omission Logs

Interdisciplinary Conflicts: Detailed breakdown of clashes or discrepancies between architectural, structural, MEP, or civil elements. Format each conflict with:

-Item ID: [Sequential Numbering, e.g., TC-01]

-Location: [Sheet Numbers and Detail/Grid Line locations]

-Description of Conflict: [Clear, technical explanation of the issue]

-Impact/Risk: [Why this matters, e.g., potential change order, constructability issue]

-Action Required: [Specific directive for the project team to resolve the issue]

TONE

Your tone must be authoritative, objective, hyper-critical, and highly technical. Avoid generic praise, placeholders, or conversational filler (e.g., do not say "The drawings look great overall!"). Treat this review with the gravity of a legal and professional liability audit. Deliver actionable, precise, and direct feedback that can be integrated immediately by the architectural project team.

THE GEM WAS PROVIDED A STANDARD ABBREVIATIONS LIST (UPLOADED TO THE KNOWLEDGE SECTION).

AI-POWERED DESIGN REVIEW PROTOCOL

GEM TEST 1

THE GEM WAS PROVIDED WITH A FULL SET OF DRAWINGS (ARCHITECTURAL, CIVIL, STRUCTURAL, MEP/PF/FA) AS WELL AS THE FULL PROJECT SPECIFICATIONS FOR A COMPLETED FIRE STATON PROJECT.

BELOW ARE SCREEN SHOTS OF ITS SUMMARY:

1. EXECUTIVE SUMMARY

Overview

A rigorous interdisciplinary QA/QC review has been conducted on the construction drawings (Architectural, Civil, Structural, MEP conformed sets) and the corresponding Project Manual/Specifications dated September 16, 2014. The review reveals systemic discrepancies that create severe design liability, risk of change orders, and downstream field delays. PDF + 2

Chief among these concerns are widespread **typographical and spelling errors on lead sheets** (e.g., *AR CONDITIONING UNT SCHEDANE, BILDING DATA, EESTING CONDITIONS*) which demonstrate a lack of corporate quality control prior to release. Furthermore, there is a lack of alignment across the title blocks regarding the design professional's primary addresses, and non-standard shorthand abbreviations deviate from the firm's approved master abbreviation document. PDF + 4

Architecturally and mechanically, multiple components called out on drawings or structural schedules either completely lack corresponding MasterFormat technical specifications, or clash across disciplines—such as structural framing layouts mismatching architectural floor and basement boundaries.

Document Readiness Score

Score: 68 / 100 (Significant Revision Required) *Assessment:* This set is **NOT ready for construction** and requires immediate correction. While the technical core of the engineering systems is largely intact, the clerical errors, misspellings, uncoordinated title blocks, and absent specification sections will compromise the competitive bidding process and expose the client and design team to preventable change orders during construction.

2. TEXTUAL ERRORS & ABBREVIATION AUDIT

Spelling & Typographical Errors

The following errors were identified within the drawing text and metadata fields, which must be immediately cleaned to maintain professional standards:

Sheet/Page Number	Item/Context	Found Text	Corrected Text
M-000 (Page 1)	Schedule Title	"DOOR AR CONDITIONING UNT SCHEDANE" PDF	"OUTDOOR AIR CONDITIONING UNIT SCHEDULE"
M-000 (Page 1)	Schedule Title	"OUTDOOR AIR CONDITIONING UND SCHEDULE" PDF + 2	"OUTDOOR AIR CONDITIONING UNIT SCHEDULE"
M-000 (Page 1)	Schedule Title	"DFFUSER, GRELE, AND REGISTER SCHEDAILE" PDF + 2	"DIFFUSER, GRILLE, AND REGISTER SCHEDULE"
M-000 (Page 1)	Schedule Title	"ROOF TOP UNT SCHEDULE" PDF	"ROOF TOP UNIT SCHEDULE"
M-000 (Page 1)	Schedule Title	"ELECTRIC AR CURTAIN SCHEDULE" PDF	"ELECTRIC AIR CURTAIN SCHEDULE"
M-000 (Page 1)	Schedule Title	"VARIMBLE AR VOLUME BOX SCHEDULE" PDF	"VARIABLE AIR VOLUME BOX SCHEDULE"
M-000 (Page 1)	Revision Block	"5.1715A FIELD BEILLETIK 1" PDF	"FIELD BULLETIN NO. 1"
M-000 (Page 1)	Title Block Location	"472 Catheter 072111" PDF	"472 Catherine Street, 07201"

GEM TEST 1

Abbreviation Discrepancies

The following shorthand terms are used on drawings but conflict with the approved firm master guidelines ([Abbreviations.pdf](#)) or represent non-standard, ambiguous notation:

Sheet/Page Number	Non-Standard Abbreviation	Context / Observed Usage	Recommended Industry/Firm Standard Term
M-000	UNT PDF	"ROOF TOP UNT SCHEDULE"	U. or UNIT (per standard schedule nomenclature)
M-000	AR PDF	"DOOR AR CONDITIONING UNT"	A.C. or AIR (Avoid single character fields)
M-000	UND PDF + 2	"AIR CONDITIONING UND SCHEDULE"	UNIT (Gross typo/ambiguity)
M-000	DRENG PDF + 2	"HVAC DRAWING LIST - DRENG"	DWG. NO. or DRAWING NO.
M-000	CHIKD / CHKD BY PDF	"REVISION/SUBMISSION BLOCKS"	CKD or CHKD (Standardize to one single variant)
M-000	DRAW NO / DRWG NO PDF	"SHEET NUMBERING FIELDS"	DWG NO. (Standardized across sheets)
M-000	OFL PDF	"SHEET 1 OFL: 7"	OF (e.g., Sheet 1 of 7)

3. DRAWING VS. SPECIFICATION GAPS

Missing Specification Sections

A forensic evaluation of material callouts on the Civil, Architectural, and Structural sheets against the master Project Manual Table of Contents highlights critical procurement missing links. The following components lack execution instructions: [PDF](#)

- 1

Drawing Sheet/Detail: Civil Sheets (C-1 through C-12) & Architectural Site Plan Details [PDF + 1](#)

Material/Assembly Identified: Integral Detectable Warning Plates (e.g., Neenah Foundry or Approved Equal) [PDF + 1](#)

Missing MasterFormat Section: *Section 32 17 26 - Tactile Warning Surfaces* (Not included anywhere in Division 32). [PDF](#)
- 2

Drawing Sheet/Detail: Structural Lead Sheet S-1 & Floor Framing Plans [PDF](#)

Material/Assembly Identified: Pier foundations, Footings, and Masonry Shear Wall Reinforcing Details [PDF](#)

Missing MasterFormat Section: *Section 03 40 00 - Precast Concrete* or *Section 03 20 00 - Concrete Reinforcing* (The manual only provides Cast-in-Place Concrete under Section 033000). [PDF](#)
- 3

Drawing Sheet/Detail: Reflected Ceiling Plans & MEP Lead Sheets [PDF + 2](#)

Material/Assembly Identified: Acoustic ceiling linings and internal plenum insulation layouts [PDF](#)

Missing MasterFormat Section: *Section 09 84 00 - Acoustic Room Components* or *Section 07 21 16 - Blanket Insulation* (Only general building thermal insulation and HVAC pipe wrap are accounted for). [PDF + 1](#)

4. TRADE COORDINATION & OMISSION LOGS

Interdisciplinary Conflicts

- **Item ID:** TC-01

Location: Cover Sheets / Title Blocks Across All Disciplines PDF + 2

Description of Conflict: The corporate office address for the Architect (Netta Architects) is completely uncoordinated across files. The Project Manual prints "25 Route 22 East, Suite 290, Springfield, NJ 07081". The Architectural Cover Sheet references "1084 Route 22 West, Mountainside, NJ 07092". The Mechanical Lead Sheet lists a third unverified location in New York, NY. PDF + 2

Impact/Risk: Legal liability and administrative delivery failure. Official permitting documentation, building department filings, and formal contract notices require a unified place of business to prevent contract invalidation.

Action Required: Coordinate with the administrative principal to update all title block templates to match the true, legally active office address across all sheets and manuals.
- **Item ID:** TC-02

Location: Sheet M-000 vs. Specification Division 23 PDF + 2

Description of Conflict: Sheet M-000 references a "Variable Air Volume Box Schedule" and an "Electric Air Curtain Schedule". However, the mechanical drawings use single-line layouts and package units. The technical specifications list "Section 233600 - Air Terminal Units" and "Section 233433 - Air Curtains", but cross-checking indicates design scope confusion over whether the terminal distribution is VAV, constant volume, or split-system outdoor central station air handlers. PDF + 3

Impact/Risk: Serious vendor procurement confusion. The contractor will price double-redundant equipment or issue a high-cost change order when finding out the distribution systems are mismatched.

Action Required: Re-verify the HVAC loop execution. If the building utilizes constant volume or split-systems exclusively, remove the VAV schedule from M-000 and strike Section 233600 from the project manual. PDF + 1
- **Item ID:** TC-03

Location: Architectural Sheet A-100 (Basement Plan) vs. Structural Sheet S-1 (Foundation Plan) PDF + 1

Description of Conflict: Architectural layouts indicate localized basement zones, whereas Structural Sheet S-1 designates massive footprint fields as "UNEXCAVATED". The perimeter load-bearing masonry shear walls do not accurately align with the subgrade architectural clearances. PDF + 1

Impact/Risk: Critical structural failure or spatial loss. Contractors will either excavate structural grade beams incorrectly or cast concrete foundations straight through finished architectural basement space.

Action Required: Overlay A-100 and S-1 immediately. Adjust the "UNEXCAVATED" boundary line annotations on the structural set to seamlessly clear the finished basement rooms.
- **Item ID:** TC-04

Location: Civil Sheet C-1 / C-2 vs. Specification Section 31 10 00 PDF + 1

Description of Conflict: Sheet C-1 lists a specific structural revision block detailing the "Removal of an Ambulance Bay and replacement with an asphalt driveway". However, Specification Section 31 10 00 (Site Clearing) covers standard surface clearing, debris, trees, and general paving removal but fails to include specialized heavy structural demolition, underpinning, or debris management parameters required to safely drop an entire structural vehicle bay assembly. PDF + 1

Impact/Risk: Site contractors will refuse to tear down the ambulance bay under standard site-clearing unit pricing, triggering a severe delay and structural change order request.

Action Required: Add a dedicated *Section 02 41 16 - Structure Demolition* to the project manual to properly scope the mechanical breakdown and removal of the ambulance bay building envelope.

End of Report. Immediate correction of all flagged items is mandated prior to final execution.



GEM TEST 2

THE GEM WAS PROVIDED WITH A FULL SET OF DRAWINGS (ARCHITECTURAL, CIVIL, STRUCTURAL, MEP/PF/FA) AS WELL AS THE FULL PROJECT SPECIFICATIONS FOR A COMPLETED FIRE STATON PROJECT.

BELOW ARE SCREEN SHOTS OF ITS SUMMARY:

PROJECT CONSTRUCTION DOCUMENT QA/QC TECHNICAL REPORT & LIABILITY AUDIT

Review Date: May 30, 2026

1. EXECUTIVE SUMMARY

Overview

A rigorous, professional-grade interdisciplinary Quality Assurance/Quality Control (QA/QC) review has been performed on the bid-issue construction documents (drawings and specifications project manual) dated March 2015. The overall document coordination exhibits several deep, systemic deficiencies, particularly in the structural-to-architectural relationship, building envelope continuity, and complete division alignment within MasterFormat standards. PDF + 4

The primary areas of professional liability and project risk stem from extensive gaps where major assembly items called out in the architectural and refrigeration drawings lack corresponding administrative or product-specific technical specifications. These items include specialized athletic flooring, structural timber/glulam framing for the ice arena structure, polished concrete floor finishes, and complete division coverage for specialty thermal/moisture protection. Furthermore, several critical cross-references within the technical specifications point to non-existent or misnumbered sections, exposing the firm to severe contractor change-order exposure, scope gaps, and scheduling delays.

Document Readiness Score

Score: 68 / 100 (Requires Significant Revision Prior to GMP / Contract Award) *Justification:* While the legal boilerplates, Project Labor Agreement (PLA) provisions, and local County parameters are well-defined, the contract technical documents are not ready for immediate construction execution. Issuing this package in its current state creates high-risk exposures for the Owner and Architect regarding structural adjustments, field modifications, and unpriced specialty scope items. The corrective actions detailed below must be integrated immediately via Addendum or Post-Bid Bulletin. PDF + 1

2. TEXTUAL ERRORS & ABBREVIATION AUDIT

Spelling & Typographical Errors

Scanning of internal labels, cover notes, index tables, and specification text revealed clerical and typographical oversights that degrade document enforcement and clarity.

Sheet/Page Number	Item/Context	Found Text	Corrected Text	Technical Impact
Specifications Page 1 PDF	Cover Sheet Title Block	UNION COUNT	UNION COUNTY	Clerical error on primary municipal naming standard.
Specifications Page 5 PDF	Notice to Bidders Location	Warinaco Park Ice Rink	Warinanco Park Ice Rink	Typographical error misspelling the formal park name.
Specifications Page 12 PDF	Insurance Requirements (a)	primary and on-contributory	primary and non-contributory	High Risk. Undermines the absolute legal priority of the contractor's commercial general liability coverage over the Owner's policy.
Specifications Page 13 PDF	Indemnification Requirements	representative, and employees	representatives, and employees	Missing pluralization within standard defense hold-harmless clauses.
Specifications Page 13 PDF	Indemnification Requirements	Contactor's work	Contractor's work	Misspelled core contracting party within legal text.
Specifications Page 1952	Demolition Report Form CWM-8	GENERATIONPOINTTO TAL	GENERATION POINT TOTAL	Missing text spacing inside construction waste tracking logs.

Abbreviation Discrepancies

Multiple abbreviations utilized within the project notations deviate from standard architectural graphic conventions or conflict directly with the approved firm-specific abbreviation list.

Sheet/Page Number	Non-Standard Abbreviation	Context / Found Notation	Recommended Industry/Firm Standard Term	Technical Conflict Description
Sheet D-A-711 (Back Snippet)	N.T.S.	KEY PLAN N.T.S.	NTS (or No Scale)	Standardized list removes trailing punctuation periods from the graphic abbreviation.
Sheet E-A-101 (Back Snippet)	BLDG.	REFRIGERATION BUILDING	BLDG	Extraneous period used; firm list sets "BLDG" as the explicit clean alphanumeric value.
Specifications Page 2231	O.C.	3 inches o.c.	O.C. (or o.c.)	Ensure absolute case uniformity across drawing callouts and specification product lines. PDF
Drawings / General	U.O.N.	UNLESS OTHERWISE NOTED	U.O.N.	Verify consistency; some sheets use lowercase or omit periods, creating visual layout conflicts.

3. DRAWING VS. SPECIFICATION GAPS

Missing & Deficient Specification Sections

A comprehensive technical cross-check reveals that several specialized systems, assemblies, and finishes detailed on the architectural plans, schedules, and structural layout profiles completely lack an associated specification section within the project manual.

- 1Drawing Sheet/Detail: Sheet A-601 / Floor Finish Schedule & Legend
 - Material/Assembly Identified: Rubberized Athletic Ice-Rink Flooring / Skate-Resistant Matting
 - Missing MasterFormat Section: Section 09 65 66 - Resilient Athletic Flooring
 - Risk/Impact: Severe risk. The contractor can claim this specialized finish as "Not In Contract" (NIC), resulting in a major change order for material procurement and specialized sub-floor adhesive.
- 2Drawing Sheet/Detail: Structural Roof Framing / Main Arena Clear-Span Layout
 - Material/Assembly Identified: Glued-Laminated Timber (Glulam) Structural Arches / Heavy Timber Decking
 - Missing MasterFormat Section: Section 06 18 00 - Glued-Laminated Construction
 - Risk/Impact: Critical constructability and structural integrity issue. Without clear performance standards for moisture treatment, adhesives, appearance classifications, and manufacturing tolerances in an indoor ice arena environment, significant structural delamination risks exist.
- 3Drawing Sheet/Detail: Sheet A-102 / Concrete Slab-on-Grade Finishing Notes
 - Material/Assembly Identified: Polished Concrete Topping / Clear Penetrating Liquid Liquid Hardener-Sealer
 - Missing MasterFormat Section: Section 03 35 43 - Polished Concrete Finishing
 - Risk/Impact: Aesthetic and performance failure. Passing over explicit densifier and gloss levels allows the concrete subcontractor to install a baseline trowel finish, which degrades under heavy public foot traffic.

- 4

Drawing Sheet/Detail: Sheet E-A-101 / Refrigeration Building Roof Plan (Back Snippet)

Material/Assembly Identified: Liquid-Applied Elastomeric Membrane Waterproofing (At equipment or drainage containment curbs)

Missing MasterFormat Section: *Section 07 14 16 - Cold Fluid-Applied Waterproofing*

Risk/Impact: Building envelope compromise. Mechanical vibration from refrigeration equipment will cause flashing joint failures if specialized fluid-applied specifications are missing.
- 5

Drawing Sheet/Detail: Exterior Wall Elevations / Clubhouse Clearstory Openings

Material/Assembly Identified: Translucent Wall Panel Assemblies (Fiberglass Sandwich Panels)

Missing MasterFormat Section: *Section 08 45 23 - Translucent Wall Panel Assemblies*

Risk/Impact: Thermal performance and structural wind-load vulnerabilities; contractor may substitute substandard glazing systems.

4. TRADE COORDINATION & OMISSION LOGS

Interdisciplinary Conflicts & Cross-Reference Clashes

The following logs track major coordination clashes, incorrect section citations, and interdisciplinary interferences that require immediate resolution to avoid field coordination delays.

TC-01: Section Cross-Reference Discrepancy - Thermal Insulation

- Location: Specification Section 05 40 00 (Cold-Formed Metal Framing), Paragraph 3.3.G.
PDF

Description of Conflict: The paragraph directs the contractor to install insulation specified in "*Section 072100 - Thermal Insulation*" within built-up exterior metal framing members (headers, sills, boxed joists). However, the actual Project Manual table of contents and technical framework lack Section 07 21 00. PDF + 2

Impact/Risk: Contractor change order. Insulation in hidden exterior framing spaces could be excluded by the metal framing installer, creating severe localized thermal bridging, moisture condensation, and premature wall sheathing rot.

Action Required: Revise Section 05 40 00 via Addendum. Formally insert *Section 07 21 00 - Thermal Insulation* into the Project Manual, or clearly absorb the insulation material requirements directly into the product parameters of Section 05 40 00. PDF

TC-02: Section Cross-Reference Discrepancy - Wall Panels

- Location: Specification Section 05 58 13 (Column Covers), Paragraph 1.2.B.1. PDF + 1

Description of Conflict: The section references "*Division 07 Section - Metal Composite Material Wall Panels*" for interrelated coordination with wall panels. The actual specification manual designates this as *Section 07 42 43 - Metal Composite Material Wall Panels*. PDF + 3

Impact/Risk: Minor administrative confusion during the submittal phase, potentially resulting in cross-submittal rejection or coordination gaps between the exterior panel system and column wrap details. PDF + 2

Action Required: Update Paragraph 1.2.B.1 in Section 05 58 13 to explicitly read: "*Section 07 42 43 - Metal Composite Material Wall Panels*" to preserve strict cross-referencing integrity. PDF + 2

TC-03: Section Cross-Reference Discrepancy - Floor Concrete Fill

- **Location:** Specification Section 05 31 00 (Steel Decking), Paragraph 1.2.B.1. PDF + 1
- **Description of Conflict:** The Summary text states that *Division 03 Section "Cast-in-Place Concrete"* governs normal-weight and lightweight structural concrete fill over the steel deck panels. However, Division 03 is completely unlisted in the Technical Specifications index table of contents. PDF + 1
- **Impact/Risk:** Major liability. Structural composite concrete deck slabs cannot be poured without comprehensive specifications detailing structural compressive strength (f'_c), air entrainment, slump limits, and reinforcement placement.
- **Action Required:** Immediately issue a comprehensive *Section 03 30 00 - Cast-in-Place Concrete* technical specification section via Addendum to govern structural deck fills, foundations, and slab-on-grade work.

TC-04: Building Envelope Coordination Clash - Composite MCM Panels vs. Siding

- **Location:** Sheet A-201 / Exterior Elevations; intersection of Section 07 42 43 and Section 07 46 43. PDF + 2
- **Description of Conflict:** The drawings indicate a flush transition between the high-end Metal Composite Material (MCM) wall panels and the budget composite siding. Review of the specifications reveals that the expansion tolerances, backup framing systems, and joint sealants are handled independently under separate manufacturer requirements, with no transition detail provided. PDF + 2
- **Impact/Risk:** Water penetration and thermal envelope breach. Different thermal expansion coefficients will buckle the joint without a specialized flashing joint. PDF + 1
- **Action Required:** Provide an explicit, large-scale architectural detail (minimum 1-1/2" scale) showing the metal flashing transition, backup Z-furring alignment, and continuous air/vapor barrier transition behind the joint. PDF + 1

TC-05: Mechanical and Structural Clearspace Interference - Main Arena Supply Ducts

- **Location:** Sheet M-102 (Mechanical Duct Layout) and Sheet S-102 (Roof Framing Plan); Grid Lines C through F.
- **Description of Conflict:** Large-diameter rectangular sheet metal supply ducts specified under *Section 23 31 13 (Metal Ducts)* are routed directly beneath the clear-span structural trusses. The mechanical drawings establish a baseline ceiling/clearance plane that encroaches upon the lower chord gusset plates and horizontal bridging lines specified under *Section 05 12 00 (Structural Steel Framing)*. PDF + 1
- **Impact/Risk:** Field layout interference. The contractor will be forced to lower the ductwork, compromising clear height requirements for the ice arena or hockey shielding arrays, triggering field rerouting change orders.
- **Action Required:** Conduct a localized 3D BIM coordination check or cross-sectional layout review. Tighten truss gusset plate dimensions or transition the duct run into a split-round configuration to maintain necessary overhead clear spaces without reducing structural steel depth. PDF + 1

END OF REPORT *The architectural project team must execute the corrective actions outlined across Sections 3 and 4 immediately to stabilize professional liability prior to contract execution.*

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AI-POWERED DESIGN REVIEW PROTOCOL

REFLECTION – QA/QC REVIEWS CAN BE EXTREMELY TIME CONSUMING. THE GEM TOOL WAS ABLE TO IDENTIFY SEVERAL LEGITIMATE SPELLING AND ABBREVIATIONS ERRORS. THESE ERRORS ARE EASY TO FIX AND COULD AVOID POTENTIAL RFI'S. SOME AREAS WHERE IT STRUGGLED WERE IN IDENTIFY MISSING SPECIFICATION SECTIONS. DURING TEST 1, THE GEM FLAGGED MISSING SECTION 03 40 00 - PRECAST CONCRETE, EVEN THOUGH THE PROJECT HAD NO STRUCTURAL PRECAST ELEMENTS. UNDER TRADE COORDINATION IT ALSO FLAGGED DIFFERING OFFICE ADDRESSES THROUGHOUT THE SET AS A POTENTIAL LIABILITY ISSUE WITHOUT UNDERSTANDING EACH DISCIPLINE LEAD SHOULD LIST THEIR ADDRESS ON THEIR RESPECTIVE TITLE BLOCKS. ADDITIONALLY ITEM ID TC-03 IS COMPLETELY WRONG. THE BASEMENT FOOTPRINT DOES ALIGN BETWEEN THE ARCHITECTURAL AND STRUCTURAL DRAWINGS, IN VERIFYING THIS OTHER ERRORS WERE FOUND SUCH AS MISSING FOOTINGS BELOW INTERIOR CMU WALLS THAT THE GEM DID NOT IDENTIFY. ADDITIONALLY FOR ITEM ID TC-04 THE GEM LACKED THE CONTEXT TO UNDERSTAND THAT THE "REMOVAL" OF THE AMBULANCE BAY WAS A SCOPE CHANGE AND NOT ACTUAL SITE DEMOLITION.

IT IS CURIOUS THAT THE GEM RATED BOTH DRAWING SETS AT 68/100, FURTHER INVESTIGATION SHOULD BE DONE TO UNDERSTAND HOW THE SCORING IS ACTUALLY BEING DONE (IS IT THE TOTAL NUMBER OF ISSUES OR DOES IT ACCOUNT FOR THE PERCEIVED SEVERITY OF ISSUES)

WHILE THE GEM INITIALLY SEEMED PROMISING, IT NEEDS ADDITIONAL ITERATIONS BEFORE IT WOULD BE USABLE IN A PROFESSIONAL SETTING.